

Supply Chain Analytics Dashboard Report

Project Overview

The objective of this project was to analyze supply chain operations and develop an interactive Power BI dashboard to monitor sales performance, profitability, customer behavior, market trends, logistics efficiency, and order management.

The analysis was performed using Python, MySQL, and Power BI.

Tools Used

- Python (Pandas, NumPy)
- MySQL
- Power BI
- Excel (Data Validation & Review)

Dataset Information

Dataset: Supply Chain Dataset

Total Records: 180,519+

The dataset contains information related to:

- Orders
- Products
- Customers
- Markets
- Shipping
- Sales
- Profitability

Data Preparation

Data Cleaning

The following data cleaning activities were performed:

- Removed unnecessary columns
- Customer Email
- Customer Password
- Customer First Name
- Customer Last Name
- Customer Street
- Product Image

- Latitude
- Longitude
- Checked for missing values
- Verified data types
- Removed irrelevant fields for business analysis

Exploratory Data Analysis (EDA)

Key analyses performed:

- Sales by Product Category
 - Top Products by Sales
 - Top Products by Quantity
 - Revenue Analysis
 - Profitability Analysis
 - Delivery Performance Analysis
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Dashboard Structure

Page 1 – Executive Dashboard

KPIs

- Total Sales
- Total Profit
- Total Orders
- Late Delivery %

Visuals

- Revenue by Category
- Revenue by Market
- Monthly Sales Trend
- Delivery Status Analysis

Key Insight

The dashboard provides a high-level overview of business performance and operational efficiency.

Page 2 – Products & Profit

KPIs

- Total Products
- Product Sales

- Product Profit
- Profit Margin %

Visuals

- Top 10 Products by Sales
- Top 10 Products by Profit
- Sales vs Profit by Category
- Profit Margin by Category
- Loss Making Products

Key Insight

Fishing, Cleats, and Camping & Hiking categories generated the highest sales revenue. Product profitability varied significantly across categories, highlighting opportunities for optimization.

Page 3 – Markets & Customers

KPIs

- Total Markets
- Total Countries
- Total Customers
- Customer Segments

Visuals

- Revenue by Market
- Revenue by Country
- Top Countries by Sales
- Customer Segment Revenue
- Customer Segment Orders

Key Insight

Revenue distribution varied across global markets, helping identify high-performing regions and customer segments.

Page 4 – Logistics

KPIs

- Late Orders
- On-Time Orders
- Late Delivery %
- Shipping Modes

Visuals

- Delivery Status Analysis
- Orders by Shipping Mode
- Profit by Shipping Mode
- Average Shipping Days
- Late Deliveries by Market

Key Insight

Late deliveries represented a significant operational challenge. Shipping performance analysis highlighted opportunities to improve customer satisfaction and reduce delivery risk.

Page 5 – Orders Analysis

KPIs

- Total Orders
- Average Order Value
- Total Quantity Sold
- Order Status Types

Visuals

- Monthly Orders Trend
- Order Status Analysis
- Orders by Category
- Top 10 Cities by Orders

Key Insight

Order trends revealed demand patterns across product categories and cities, supporting better inventory and operational planning.

Business Insights

1. Fishing category generated the highest revenue among all product categories.
2. Certain products generated high sales volume but lower profit margins, indicating potential pricing or cost optimization opportunities.
3. Multiple markets contributed significantly to total revenue, emphasizing the importance of regional performance monitoring.
4. The late delivery rate exceeded 50%, indicating a need for logistics process improvement.
5. Customer purchasing behavior varied by market and segment, providing opportunities for targeted business strategies.

Conclusion

This project demonstrates end-to-end data analytics skills including:

- Data Cleaning
- Exploratory Data Analysis
- SQL Querying
- Data Modeling
- KPI Development
- Dashboard Design
- Business Insight Generation

The final dashboard enables stakeholders to monitor operational performance, identify improvement opportunities, and support data-driven decision making across supply chain operations.